TRIPOD+AI Statement Checklist

Item	Section / Topic	Checklist Item	Reported Location in Manuscript
Title			
1	Title	Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted.	**Title page:** "NeurALLNet: An attention-based spiking neural network for energy-efficient multi-class classification of acute lymphoblastic leukemia"
Abstract			
2	Abstract	See TRIPOD+AI for Abstracts checklist.	**Abstract:** Structured into Objectives, Methods, Results, and Conclusion as per editorial request.
Introduction			

3a	Background	Explain the healthcare context (including whether diagnostic or prognostic) and rationale for developing or evaluating the prediction model, including references to existing models.	**Section 1 (Introduction):** Paragraphs 1-2.
3b	Background	Describe the target population and the intended purpose of the prediction model in the context of the care pathway, including its intended users.	**Section 1 (Introduction):** Paragraph 1 (LMICs, rural clinics) & **Section 5 (Discussion):** Lab-on-a-Chip integration.
3c	Background	Describe any known health inequalities between sociodemographic groups.	**Section 1 (Introduction):** Paragraph 1 (Diagnostic delays reducing survival in LMICs vs high-income settings).
4	Objectives	Specify the study objectives, including whether the study describes the development or validation of a prediction model (or both).	**Section 1 (Introduction):** Paragraph 3 ("To address these gaps, we propose NeurALLNet...").
Methods			

5a	Data	Describe the sources of data separately for the development and evaluation datasets, the rationale for using these data, and representativeness of the data.	**Section 3.2 (Dataset & Preprocessing)** and **Section 4.2 (External Validation)**.
5b	Data	Specify the dates of the collected participant data, including start and end of participant accrual; and, if applicable, end of follow-up.	**N/A:** Not provided in the original public dataset repositories.
6a	Participants	Specify key elements of the study setting (eg, primary care, secondary care, general population) including the number and location of centres.	**Section 3.2:** Taleqani Hospital, Tehran. **Section 4.2:** Multiple hospitals in Tehran.
6b	Participants	Describe the eligibility criteria for study participants.	**Section 3.2:** 89 individuals suspected of ALL, including 25 healthy subjects.
6c	Participants	Give details of any treatments received, and how they were handled during model	**N/A:** Diagnostic imaging study; no treatments evaluated.

		development or evaluation, if relevant.	
7	Data preparation	Describe any data pre-processing and quality checking, including whether this was similar across relevant sociodemographic groups.	**Section 3.2 (Dataset & Preprocessing)** & **Section 3.4 (Training Configuration, Augmentation, and Reproducibility)**.
8a	Outcome	Clearly define the outcome that is being predicted and the time horizon, including how and when assessed, the rationale for choosing this outcome.	**Section 3.1 (Problem Formulation):** 4 discrete ALL subtypes. Time horizon N/A (diagnostic).
8b	Outcome	If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the outcome assessors.	**Section 1 (Introduction):** Ground truth established by trained hematopathologists in the original dataset.
8c	Outcome	Report any actions to blind assessment of the outcome to be predicted.	**N/A:** Retrospective dataset analysis.

9a	Predictors	Describe the choice of initial predictors and any pre-selection of predictors before model building.	**Section 3.3 (NeurALLNet Architecture):** Direct coding of static RGB images; no manual feature pre-selection.
9b	Predictors	Clearly define all predictors, including how and when they were measured.	**Section 3.1:** Continuous pixel values of peripheral blood smear RGB images.
9c	Predictors	If predictor measurement requires subjective interpretation, describe the qualifications and demographic characteristics of the predictor assessors.	**N/A:** Automated pixel extraction.
10	Sample size	Explain how the study size was arrived at, and justify that the study size was sufficient to answer the research question.	**Section 3.2:** Details train/val/test splits and resampling strategies for a balanced training set of 2,000 images.
11	Missing data	Describe how missing data were handled. Provide reasons for omitting any data.	**N/A:** No missing data in the image datasets.

12a	Analytical methods	Describe how the data were used in the analysis, including whether the data were partitioned, considering any sample size requirements.	**Section 3.2:** Stratified 80/10/10 split at the image level.
12b	Analytical methods	Depending on the type of model, describe how predictors were handled in the analyses (functional form, rescaling, transformation, or any standardisation).	**Section 3.3:** Image resizing and normalization techniques.
12c	Analytical methods	Specify the type of model, rationale, all model building steps, including any hyperparameter tuning, and method for internal validation.	**Section 3.3** & **Section 3.4 (Training Configuration, Augmentation, and Reproducibility)**.
12d	Analytical methods	Describe if and how any heterogeneity in estimates of model parameter values and model performance was handled and quantified across clusters.	**Section 4.2 (External Validation)** & **Section 5 (Discussion):** Limitations of geographical overlap acknowledged.

12e	Analytical methods	Specify all measures and plots used (and their rationale) to evaluate model performance and, if relevant, to compare multiple models.	**Section 4.1:** Accuracy, Precision, Recall, F1-score, Bootstrapped 95% CIs.
12f	Analytical methods	Describe any model updating (eg, recalibration) arising from the model evaluation.	**N/A:** No model updating performed.
12g	Analytical methods	For model evaluation, describe how the model predictions were calculated.	**Section 3.3:** Defined via Algorithm 1 (Forward Pass).
13	Class imbalance	If class imbalance methods were used, state why and how this was done.	**Section 3.2:** Two-stage resampling strategy (under-sampling majority, over-sampling minority Benign class).
14	Fairness	Describe any approaches that were used to address model fairness and their rationale.	**Section 5 (Discussion):** Limitations regarding dataset geography and need for diverse international cohorts.

15	Model output	Specify the output of the prediction model (eg, probabilities, classification).	**Section 3.3:** Softmax probabilities over discrete time steps to yield final classification.
16	Training vs evaluation	Identify any differences between the development and evaluation data in healthcare setting, eligibility criteria, outcome, and predictors.	**Section 4.2 (External Validation):** Details multicenter acquisition setup vs. primary single-center data.
17	Ethical approval	Name the institutional research board or ethics committee that approved the study and describe the participant informed consent or the ethics committee waiver of informed consent.	**Section: Ethical approval.** Formal waiver explicitly declared due to open-source, anonymized data.
18a	Open science	Give the source of funding and the role of the funders for the present study.	**Section: Funding.**
18b	Open science	Declare any conflicts of interest and financial disclosures for all authors.	**Section: Declaration of Conflicting Interests.**

18c	Open science	Indicate where the study protocol can be accessed or state that a protocol was not prepared.	**N/A:** Retrospective analysis; a priori protocol not prepared.
18d	Open science	Provide registration information for the study, or state that the study was not registered.	**N/A:** Not registered.
18e	Open science	Provide details of the availability of the study data.	**Section: Data availability statement.**
18f	Open science	Provide details of the availability of the analytical codes.	**Section: Data availability statement.** GitHub repository link provided.
19	Patient & public	Provide details of any patient and public involvement during the design, conduct, reporting, interpretation, or dissemination of the study or state no involvement.	**N/A:** No direct patient or public involvement in this retrospective computational study.
Results			

20a	Participants	Describe the flow of participants through the study, including the number of participants with and without the outcome.	**Section 3.2** & **Table 1:** Detailed class distributions before/after preprocessing.
20b	Participants	Report the characteristics overall and, where applicable, for each data source or setting.	**Section 3.2** & **Table 1.**
20c	Participants	For model evaluation, show a comparison with the development data of the distribution of important predictors.	**Section 4.2 (External Validation):** Validated on 3,242 unseen external images.
21	Model development	Specify the number of participants and outcome events in each analysis.	**Table 1.**
22	Model specification	Provide details of the full prediction model to allow predictions in new individuals and to enable third party evaluation and implementation.	**Section 3.3 (NeurALLNet Architecture)** & **Section: Data availability statement** (Open-source codebase/weights).

23a	Model performance	Report model performance estimates with confidence intervals, including for any key subgroups. Consider plots to aid presentation.	**Section 4.1:** Table 2 (Primary CI metrics), Table 3 (External CI metrics), Figure 3 & 4 (Confusion Matrices).
23b	Model performance	If examined, report results of any heterogeneity in model performance across clusters.	**Section 4.2 (External Validation):** Table 3 details specific precision/recall across external cohorts.
24	Model updating	Report the results from any model updating, including the updated model and subsequent performance.	**N/A:** No model updating performed.
Discussion			
25	Interpretation	Give an overall interpretation of the main results, including issues of fairness in the context of the objectives and previous studies.	**Section 5 (Discussion):** Paragraphs 1-2.

26	Limitations	Discuss any limitations of the study (such as a non-representative sample, sample size, overfitting, missing data) and their effects.	**Section 5 (Discussion):** Paragraph 4 (Limitations regarding geographic overlap and image-level data splits).
27a	Usability	Describe how poor quality or unavailable input data should be assessed and handled when implementing the prediction model.	**Section 5 (Discussion):** Paragraph 3 (Illumination sensitivity, IQA safeguards, and CLAHE normalization recommendations).
27b	Usability	Specify whether users will be required to interact in the handling of the input data or use of the model, and what level of expertise is required of users.	**Section 5 (Discussion):** Final paragraph (Lab-on-a-Chip integration for automated rural deployment).
27c	Future steps	Discuss any next steps for future research, with a specific view to applicability and generalisability of the model.	**Section 5 (Discussion):** Need for international cohort validation and INT8 quantization for sub-100ms CPU latency.